SYNAPSE EDGE DEFENSE

Data Loss Prevention at the Edge

Inline response inspection. Real-time blocking. Zero additional latency. Sensitive data never leaves your infrastructure unprotected.

17–28µs	Inline	5	No Agent
DECISION LATENCY	RESPONSE INSPECTION	DETECTION CATEGORIES	REQUIRED

DETECTION CAPABILITIES

Response body inspection — every category scans inline

PII — Personal Identifiable Information
SSN · Email addresses · Phone numbers
Names · Addresses · DOB · Driver's license

FIN — Financial Data
Credit card numbers (PCI) · Bank accounts
Routing numbers · IBAN · Tax IDs

SEC — Secrets & Credentials
API keys · JWT tokens · OAuth secrets
AWS keys · Private keys · SSH keys

PHI — Healthcare PHI
Medical record numbers · Health plan IDs
Diagnosis codes · Prescriptions · Lab results

API — Custom Patterns
Organization-specific regex patterns via API — internal IDs, proprietary formats, domain-specific data. Fully configurable per site.

INLINE PROTECTION FLOW

01	02	03	04	05
Client Request	WAF Inspect	Backend API	DLP Scan	Action
Attacker sends API request	Inbound detection <50µs	Processes request normally	Inspects response body	Block / Mask / Log / Alert

RESPONSE ACTIONS

Block
Replace entire response with error. Sensitive data never transmitted.

Mask
Redact matched patterns in-flight. Response delivered with data obscured.

Log
Record detection event for audit. Response passes through unmodified.

Alert
Generate real-time notification. SOC visibility without blocking traffic.

ARCHITECTURE — WHY EDGE DLP MATTERS

TRADITIONAL DLP

App ------ Cloud WAF ------ DLP Proxy ------ SIEM ------ Internet

+20-50ms added latency

Multiple hops, added latency, blind spots. Response passes through — data exfiltrated. Breach discovered 200+ days later via audit.

✗ BREACH

SYNAPSE EDGE DLP

App — Synapse WAF + DLP — Internet

17-28µs decision latency

Single hop, inline inspection, real-time blocking. 847 SSNs, 10,000 emails detected. Action: BLOCK + Alert + Actor risk +85.

✓ BLOCKED

DETECTION EXAMPLES

SOCIAL SECURITY NUMBER

```
{"user": {"ssn": "123-45-6789", "name": "John"}}
```

⚠ SSN pattern detected in response
Pattern: \b\d{3}-\d{2}-\d{4}\b
Action: Block + Log + Actor risk +25

CREDIT CARD NUMBER

```
{"payment": {"card": "4532015112830366"}}
```

⚠ Valid credit card (Luhn check passed)
Pattern: 16-digit + Luhn validation
Action: Block + Alert + PCI flag

AWS SECRET KEY

```
{"config": {"aws_secret":
"wJalrXUtnFEMI/K7MDENG..."}}
```

⚠ AWS secret access key pattern
Pattern: High entropy + known prefix
Action: Block + Critical alert + Incident

EDGE ADVANTAGES

Response Body Inspection
Not just headers — full payload analysis. JSON, XML, HTML, raw bytes.

Inline Blocking
Block before data leaves the edge. Not detect-and-alert-later.

Zero Additional Latency
DLP runs in the WAF decision path. No separate proxy hop required.

Actor Correlation
DLP violations affect actor risk scores. Cross-session behavioral memory.

COMPLIANCE SUPPORT

PCI-DSS · HIPAA · GDPR · SOC 2 · CCPA

Your data never leaves your infrastructure unprotected.
Stop data breaches at the edge — before they happen.